Polynomials — problem 10.67

A. Engel — Problem-Solving Strategies · 1998

Find all polynomials P in two variables with the following properties :
(i) For a positive integer n and all real t, x, y, $P(tx, ty) = t^n P(x, y)$.
(ii) for all real a, b, c, $P(b + c, a) + P(c + a, b) + P(a + b, c) = 0$,
(iii) $P(1, 0) = 1$ (IMO 1975).